



Programa de Doctorado en Astrofísica y Astroinformática
Centro de Investigación, Tecnología, Educación y Vinculación Astronómica
Universidad de Antofagasta

Title for a Thesis on Something Very Important About Stars, Gas, Galaxies, ...

Tesis presentada para optar al grado de
Doctor en Astrofísica y Astroinformática

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Antofagasta, 6 de noviembre de 2026

Abstract

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Keywords: star formation; Schmidt–Kennicutt law; star-forming main sequence; environment

Resumen

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Palabras clave: formación estelar; ley de Schmidt–Kennicutt; secuencia principal; entorno

Agradecimientos

Ha sido un camino largo, oscuro y lleno de deudas, pero finalmente he terminado este manuscrito. Y como la decencia y el protocolo universitario me obligan, supongo que debo dar las gracias. En primer lugar, quiero agradecer a mi profesor guía, Policarpo Avendaño —gracias, Poli, por tu apoyo constante y por no perder la paciencia conmigo durante todo este martirio académico—. Prometo que, en cuanto me paguen, te devolveré lo que te pedí prestado para ir al hipódromo. *Top, top, top.* También extiendo mis agradecimientos al Centro de Investigación, Tecnología, Educación y Vinculación Astronómica y a la Universidad de Antofagasta. Gracias por brindarme el espacio, el escritorio y los recursos para desarrollar este trabajo. El desierto resultó ser el lugar perfecto para observar el cosmos y darme cuenta de lo insignificantes que somos frente a la inmensidad del universo… y frente a mis crisis financieras. Finalmente, a mi familia y a mis amigos. Gracias por acompañarme en este pedregoso camino, por soportar mis lamentos existenciales y por no dejarme abandonado en la cuneta de la desesperanza cuando la tesis parecía no tener fin. El doctorado está listo y mi sed de conocimiento ha sido saciada. Ahora, si me disculpan, Tormenta China corre en la tercera carrera y presiento que esta vez las estrellas por fin se alinearán a mi favor.

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List of Acronyms

2MASS	Two Micron All-Sky Survey
AGN	active galactic nucleus
ALMA	Atacama Large Millimeter/submillimeter Array
IMF	initial mass function
ISM	interstellar medium
ΛCDM	Lambda cold dark matter
SFMS	star-forming main sequence

Chapter 1

Introduction

This chapter has two roles. Section 1.1 is a sample of how running text looks, with citations, acronyms, an equation and a figure. The subsequent sections form a short user manual of the template: they show, with live output, how to cite, cross-reference, insert figures and tables, write numbers and units, and use acronyms. Section 1.8 lists what to remove before submitting.

1.1 Text example

The connection between the gas content of a galaxy and its star formation rate (SFR) was first formulated by Schmidt (1959) and later calibrated for entire galaxies by Kennicutt (1998). For reviews, see (e.g., Kennicutt & Evans 2012). In the star-forming main sequence (SFMS) the SFR correlates with the stellar mass (Brinchmann et al. 2004; Speagle et al. 2014); galaxies hosting an active galactic nucleus (AGN) usually fall below it.

The Schmidt–Kennicutt relation reads

$$\Sigma_{\mathrm{SFR}} = A\,\Sigma_{\mathrm{gas}}^{N}, \tag{1.1}$$

where $N \approx 1.4$ (Kennicutt 1998); see Fig. 1.1 for a schematic illustration of Eq. (1.1).

1.2 Getting started

1. In `main.tex` edit the thesis data: title, author, supervisors, keywords). They feed the title page and the PDF metadata.
2. Replace the text of `frontmatter/abstract.tex` (English), `frontmatter/resumen.tex` and `frontmatter/agradecimientos.tex` (Spanish).
3. Write each chapter in its own file, `chapters/chapter1.tex`, `chapter2.tex`…, and put its figures in `figures/chapter1/`, `figures/chapter2/`… Appendices go in `appendices/`.

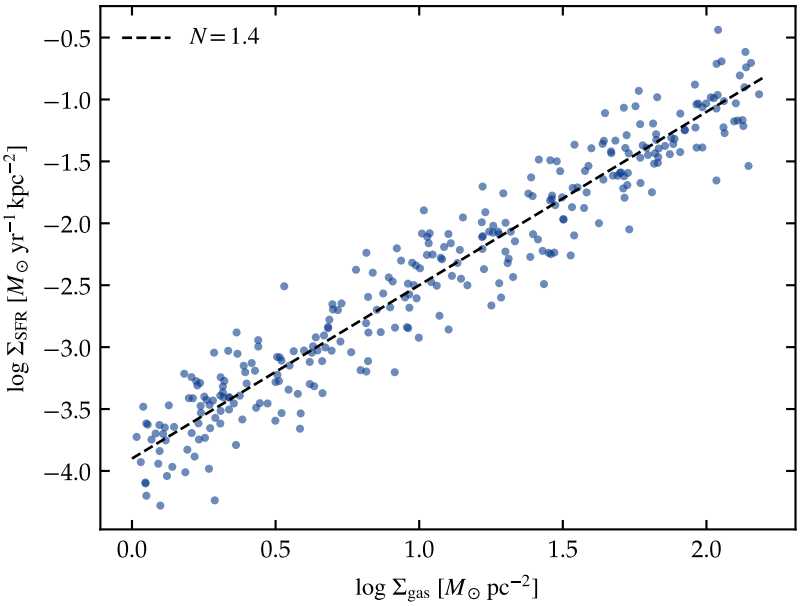


Figure 1.1: Schmidt–Kennicutt relation for mock data. The dashed line shows $N = 1.4$.

4. Add the references you cite to `references.bib` (Section 1.3).
5. Define your acronyms in `acronyms.tex` (Section 1.7).
6. Compile with XeLaTeX (see Appendix B).

To add a chapter, create `chapters/chapter6.tex` and `figures/chapter6/`, start the new file with

```
\chapter{Title}\label{ch:title}
\chapterfigs{chapter6}
```

and add `\include{chapters/chapter6}` in `main.tex`.

1.3 Citations and references

Add the BibTeX entry to `references.bib`, then cite it with its key:

- `\citet{Kennicutt1998}` → Kennicutt (1998) (author in the sentence);
- `\citep{Kennicutt1998}` → (Kennicutt 1998) (in parentheses);
- `\citep{Schmidt1959, Kennicutt1998}` → (Schmidt 1959; Kennicutt 1998);
- `\citep[e.g.,][Sect.~2]{Speagle2014}` → (e.g., Speagle et al. 2014, Sect. 2);
- `\citeauthor{Boquien2019}` and `\citeyear{Boquien2019}` → Boquien et al. (2019).

1.4 Cross-references with `\cref`

The usual LaTeX command `\ref` prints only the number, so you have to type the word yourself and remember the non-breaking space: `Figure~\ref{fig:ks}` gives Figure 1.1. This template also loads the package `cleveref`, whose command `\cref` writes the word for you, according to the type of the label.

If you type `\cref{fig:ks}` you get Fig. 1.1; for a table, `\cref{tab:sample}` gives Table 2.1; for an equation, `\cref{eq:ks}` gives Eq. (1.1); for a chapter, `\cref{ch:data}` gives Chapter 2; and for a section, `\cref{sec:cites}` gives Section 1.3;

It handles lists and ranges: `\cref{sec:sflaws,sec:cites}` gives Sections 1.1 and 1.3, and `\crefrange{sec:sflaws}{sec:xref}` gives Sections 1.1 to 1.4; the plural is automatic. `\ref` is still available and is useful when you want to write the text yourself.

1.5 Figures

Save the figure in the folder of the chapter (`figures/chapter1/`); `\chapterfigs{chapter1}` at the top of the chapter already points there, so you write only the file name, without path or extension:

```
\begin{figure}[tbp]
  \centering
  \includegraphics[width=0.7\textwidth]{ks_law_mock}
  \caption[Short title]{Long caption below the figure.}
  \label{fig:ks}
\end{figure}
```

The optional text in square brackets after `\caption` is the short title used in the list of figures. Figures made of several panels use `subfigure`, see Chapter 3 (Fig. 3.1). `[tbp]` lets LaTeX place the figure at the top, on its own page or at the bottom; avoid forcing the position.

1.6 Units, numbers and astronomy macros

Use `siunitx` for numbers and units, so that spacing is always correct: a galaxy with $M_\star = 3.2 \times 10^{10}\,\mathrm{M}_\odot$ at a distance of 12.5 Mpc, an aperture of 1.5″, or a range of 1–10 Gyr. In ranges the unit is written only once (`\qtyrange{1}{10}{\giga\year}` gives 1–10 Gyr, not “1 Gyr–10 Gyr”); this is set by the option `range-units = single` in `preamble.tex`.

Running text uses old-style figures (1959, 2012), while numbers that are measured or compared use lining (equal-height) figures: `siunitx` quantities, tables and also the digits in math

mode, which are taken from Caslon so that $-0.6 < [\mathrm{Fe/H}] < -0.4$ or $N = 1.4$ match 1.4 and 12.5 Mpc. Letters and symbols in math stay Latin Modern, which has proper math spacing. Write numbers that carry a value in math or with `\num`/`\qty`, and years or counts in plain text. Common macros are defined in `macros.tex`, where you can add your own: Hα, Hβ, [N II], [O III], H I, H II, [Fe/H], [α/Fe], $\mathrm{M}_\odot$, and also `\Mstar`, `\SFR`, `\sSFR`, `\Sgas`, `\Ssfr`.

1.7 Acronyms

Acronyms are handled by the package `glossaries-extra`. Define each acronym once, in the file `acronyms.tex`, for example

```
\newacronym{ISM}{ISM}{interstellar medium}
\newacronym[longplural={active galactic nuclei}]
           {AGN}{AGN}{active galactic nucleus}
```

and write it in the text with its label. The first time it appears the long form is spelled out; afterwards only the short form is printed. You never have to remember whether you have already introduced it.

- `\gls{ISM}` → interstellar medium (ISM) (first use), and then `\gls{ISM}` → ISM;
- `\glspl{AGN}` → active galactic nuclei (AGNs) (plural, first use; afterwards AGNs; irregular plurals are set with `longplural` as above);
- `\Gls{IMF}` → Initial mass function (IMF) (capital first letter, at the start of a sentence; `\Glspl` for the plural);
- `\glsxtrshort{ALMA}` → ALMA and `\glsxtrlong{ALMA}` → Atacama Large Millimeter/submillimeter Array (always short / always long, without changing the first-use state);
- `\gls{LCDM}` → Lambda cold dark matter (ΛCDM) (math is allowed in the short form; add `sort={LCDM}` so it is sorted correctly in the list).

For surveys and instruments, astronomy papers usually give the reference at the first mention. `\glscite{label}{bibkey}` does it automatically: `\glscite{2MASS}{Skrutskie2006}` gives "Two Micron All-Sky Survey (2MASS; Skrutskie et al. 2006)" the first time, and "2MASS" afterwards. An optional third argument is passed to `\citep` as a post-note: `\glscite{2MASS}{Skrutskie2006}[Sect.~2]`.

Good practice

- **Only the acronyms you use appear in the List of Acronyms** (after the List of Tables, also in the table of contents), sorted alphabetically. You can keep definitions in `acronyms.tex`

that you do not use yet.

- **In captions, section titles, the Abstract and the Resumen use `\glsxtrshort`** (or plain text): `\gls` there would "use up" the first expansion, which would then happen in the list of figures or in the front matter instead of in the text.
- **First-use state is reset at the start of Chapter 1** (`\glsresetall` after `\mainmatter` in `main.tex`). To spell all acronyms out again at the beginning of every chapter, add `\glsresetall` right after each `\chapter{...}`; to reset a single one, `\glsreset{ISM}`.
- **Labels are case sensitive** and should contain only letters and digits.
- The list is built by `makeglossaries`, which `latexmk` and Overleaf run automatically (see Appendix B). If the list is empty or outdated, compile once more.
- Page numbers are not shown in the list; to show where each acronym is used, delete `nonumberlist` in the `glossaries-extra` options of `preamble.tex`. To make acronyms clickable links to the list, delete `nohypertypes={acronym}`.

1.8 Before submitting: checklist

1. Delete Sections 1.2 to 1.8 of this chapter (the user manual) and the example text, figures and tables of Chapters 1 to 3 and Appendices A and B; delete the corresponding `figures/` files.
2. Remove the example entries you do not cite from `references.bib` (unused entries are not printed, but they make the file harder to maintain) and the example acronyms you do not need from `acronyms.tex`.
3. Check that no `\todo{...}` notes remain (they are printed in red).
4. Comment out `\includeonly` in `main.tex` and compile the full document from scratch, so that all references, citations and the lists are up to date (no "??" in the PDF).

Chapter 2

Data and Methods

2.1 Sample selection

The sample (Fig. 2.1, Table 2.1) is defined by stellar mass and SFR, estimated with CIGALE (Boquien et al. 2019).

2.2 Tables

Tables use the thesis style defined in `preamble.tex`. Three environments are available: `uatblr` (plain), `uatalltblr` (with table notes, used inside `table`) and `ualongtblr` (multi-page, see Appendix A). They are built on `tabularray`, so every `tabularray` key still works: `column{1}={font=\bfseries}` for a bold first column, `row{even}={bg=tablegray}` for zebra rows, `Q[si={...}]` for decimal alignment, and `\TblrNote` for notes. In Table 2.1 the header

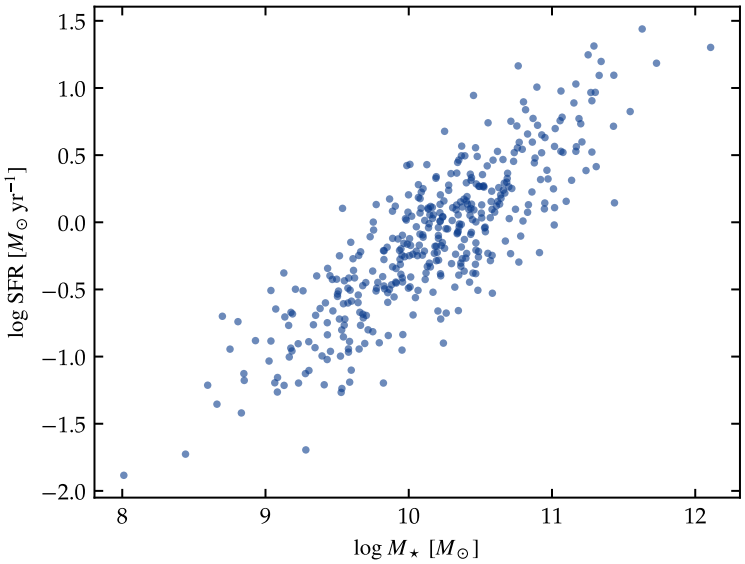


Figure 2.1: Distribution of the (mock) sample in the $M_\star$–SFR plane.

Table 2.1: Properties of the example sample (mock values).

ID	z	$\log M_\star$ [a] ($M_\odot$)	$\log \mathrm{sSFR}$ [b] (1/yr)	**Environment**
G001	0.0123	10.35	−9.80	Cluster
G002	0.0210	9.87	−9.41	Field
G003	0.0187	10.92	−10.25	Cluster
G004	0.0305	10.10	−9.55	Group
Mean	0.0206	10.31	−9.75	—

[a] Stellar mass derived from SED fitting.
[b] Specific SFR averaged over the last 100 Myr.
Notes: All values are for illustration only.

has two rows (names and units) and the last row is a summary row.

2.3 Equations

Equations are numbered by chapter; give each one a label and refer to it with `\cref{eq:sfms}` (Section 1.4). This is an example: the main sequence is commonly parametrized as

$$\log \mathrm{SFR} = \alpha \, \log M_\star + \beta, \tag{2.1}$$

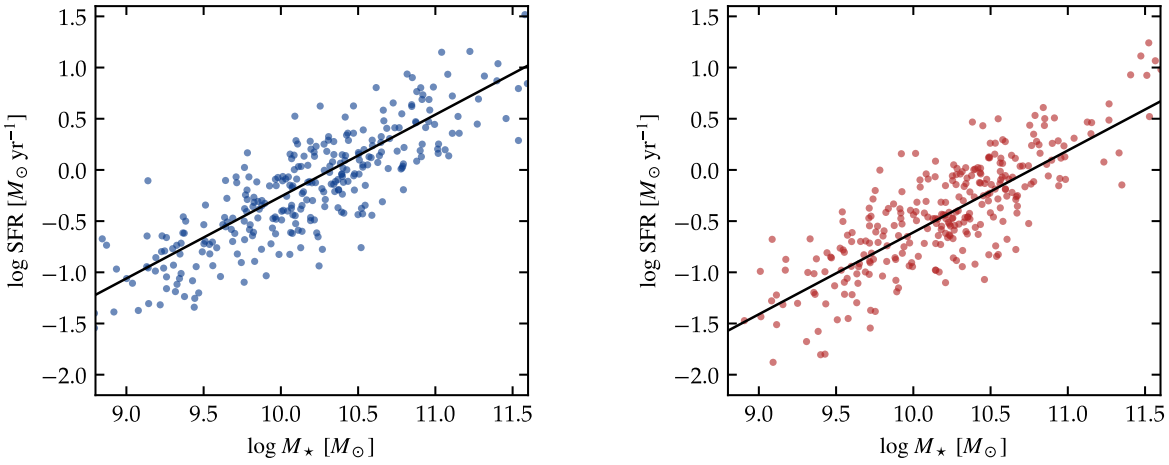
with slope α and intercept β (Speagle et al. 2014). Specific SFR is $\mathrm{sSFR} = \mathrm{SFR}/M_\star$.

Chapter 3

Results

3.1 Some plots

Figure 3.1 uses the `subfigure` environment (package `subcaption`): each panel has its own `\caption` and `\label`, and the main caption describes the whole figure. Panels can be referenced individually: Fig. 3.1b shows a lower normalisation, as reported previously (e.g., Brinchmann et al. 2004).



(a) Field galaxies.

(b) Cluster galaxies.

Figure 3.1: Star-forming main sequence (mock data) for (a) field and (b) cluster galaxies. Solid lines are linear fits.

Chapter 4

Discussion

4.1 Comparison with the literature

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Chapter 5

Conclusions and Future Work

5.1 Conclusions

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5.2 Future work

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Appendix A

Galaxy catalogue

Table A.1 is a large table that continues over several pages. It is set on landscape pages. Numbers are right-aligned, so give them the same number of decimals in each column.

Table A.1: Catalogue of the example sample (mock values).

ID	RA (deg)	Dec (deg)	z	$\log M_\star$	$\log \mathrm{SFR}$	$\log \mathrm{sSFR}$	$\log \Sigma_{\mathrm{gas}}$	Environment	Comments
G001	118.341	−51.703	0.0386	9.17	0.00	−9.17	1.19	Cluster	Edge-on
G002	18.123	−36.149	0.0113	9.22	−0.31	−9.53	1.92	Cluster	
G003	225.719	−27.935	0.0109	10.41	−1.36	−11.77	0.95	Field	AGN candidate
G004	106.363	−52.066	0.0135	9.74	0.79	−8.96	0.89	Field	Edge-on
G005	135.339	−29.874	0.0110	9.14	−0.92	−10.07	1.69	Group	
G006	167.961	−9.211	0.0250	9.60	−1.00	−10.59	1.85	Cluster	Blended in WISE
G007	188.819	−11.867	0.0423	9.69	1.24	−8.45	0.79	Group	
G008	269.999	−51.641	0.0310	9.09	0.37	−8.72	1.82	Field	
G009	124.043	−40.740	0.0313	10.91	−1.31	−12.22	0.75	Group	
G010	248.965	−56.425	0.0424	9.74	0.12	−9.62	1.69	Group	Blended in WISE
G011	255.820	−11.213	0.0243	11.26	−0.50	−11.76	1.58	Group	
G012	81.373	−44.191	0.0427	9.95	1.07	−8.89	1.39	Cluster	
G013	145.575	−44.719	0.0144	10.03	0.04	−9.99	1.73	Group	
G014	340.206	−51.699	0.0163	9.56	−0.85	−10.40	1.38	Field	
G015	96.961	−59.775	0.0277	9.89	0.09	−9.80	2.12	Field	
G016	164.825	−12.096	0.0527	10.63	0.07	−10.57	1.24	Group	Merger remnant
G017	173.533	−37.976	0.0170	11.36	−0.27	−11.63	0.78	Field	
G018	40.833	−28.827	0.0332	11.28	0.22	−11.06	0.71	Cluster	
G019	56.993	−46.126	0.0243	9.87	−1.16	−11.03	1.96	Group	
G020	174.342	−55.276	0.0128	9.82	−0.76	−10.58	1.93	Cluster	
G021	76.825	−7.639	0.0250	10.66	1.06	−9.60	1.81	Group	Close pair
G022	248.669	−45.639	0.0252	9.40	0.66	−8.74	1.45	Field	
G023	227.755	−26.272	0.0451	10.82	−0.95	−11.77	0.98	Group	
G024	74.971	−32.897	0.0424	11.38	0.71	−10.66	1.36	Cluster	Poor SED fit
G025	161.530	−8.464	0.0544	11.29	−0.48	−11.77	0.95	Cluster	
G026	73.847	−48.759	0.0373	11.16	0.85	−10.31	1.37	Field	Poor SED fit
G027	284.875	−55.337	0.0390	11.18	0.69	−10.49	1.80	Group	
G028	156.874	−25.029	0.0121	11.27	0.52	−10.75	1.34	Field	Close pair

Continued on next page

Table A.1 (continued)

ID	RA (deg)	Dec (deg)	z	$\log M_\star$	$\log \mathrm{SFR}$	$\log \mathrm{sSFR}$	$\log \Sigma_{\mathrm{gas}}$	Environment	Comments
G029	258.680	−50.650	0.0140	9.36	1.03	−8.33	1.89	Cluster	
G030	235.044	−40.728	0.0338	9.31	−1.46	−10.77	2.15	Field	Merger remnant
G031	189.303	−8.651	0.0284	11.09	0.81	−10.28	0.94	Group	Edge-on
G032	107.538	−46.770	0.0356	9.62	−0.33	−9.95	0.81	Field	Poor SED fit
G033	319.196	−23.564	0.0463	10.24	0.82	−9.42	2.01	Cluster	AGN candidate
G034	188.227	−58.971	0.0287	9.44	−1.49	−10.93	1.88	Cluster	AGN candidate
G035	170.723	−20.114	0.0342	9.78	−0.05	−9.83	1.49	Cluster	
G036	91.973	−44.770	0.0443	10.22	0.07	−10.15	1.82	Cluster	
G037	118.965	−6.465	0.0365	9.48	−0.72	−10.20	1.41	Group	
G038	249.726	−11.791	0.0523	9.62	0.07	−9.56	2.11	Group	AGN candidate
G039	150.823	−38.420	0.0229	10.61	−0.30	−10.91	0.94	Group	Merger remnant
G040	318.959	−51.505	0.0417	10.58	−1.10	−11.68	2.01	Group	
G041	266.339	−54.823	0.0496	9.39	0.37	−9.02	0.96	Field	Compact starburst
G042	352.925	−37.790	0.0278	9.86	−1.24	−11.10	1.19	Group	
G043	159.160	−59.005	0.0236	10.50	−0.07	−10.56	0.70	Cluster	Merger remnant
G044	34.421	−45.044	0.0506	9.44	0.62	−8.82	1.91	Field	Tidal features
G045	147.082	−30.487	0.0322	10.19	−0.58	−10.77	1.05	Field	

Notes: Coordinates in J2000. All values are for illustration only.

Appendix B

Compiling the document

This template must be compiled with **XeLaTeX** (not pdfLaTeX), because it uses OpenType fonts and Unicode. The output is a PDF. Besides XeLaTeX, two helper programs are run: **BibTeX** for the bibliography and **makeglossaries** for the List of Acronyms.

B.1 On your computer.

With a full TeX Live or MiKTeX installation, run in the folder of the thesis

```
latexmk
```

It reads the file `latexmkrc`, uses XeLaTeX and runs BibTeX and `makeglossaries` as many times as needed. `latexmk -c` deletes the auxiliary files. Without `latexmk`, the full sequence is

```
xelatex main
bibtex main
makeglossaries main
xelatex main
xelatex main
```

`makeglossaries` is a Perl script; on Windows without Perl use `makeglossaries-lite main` instead. In any case the package option `automake` (in `preamble.tex`) builds the list by itself during compilation, so it usually works even if you forget this step.

B.2 On Overleaf.

Upload the zip (*New Project* → *Upload Project*). Then open the project **Menu** (top left) → **Settings** and change **Compiler** to **XeLaTeX**; the default compiler (pdfLaTeX) fails with this template. Overleaf runs BibTeX and `makeglossaries` automatically. If the output looks outdated or strange, use *Recompile from scratch* to clear old auxiliary files. Free accounts have a compilation time limit that a large thesis can exceed; the tips below reduce the time.

B.3 Overleaf on your own computer (free).

Overleaf is open source: the Community Edition can be installed and run locally, so you do not need a paid plan. The code is at github.com/overleaf/overleaf; the usual way to run it is with Docker, following the instructions of the Overleaf Toolkit. Once it is running you use it in the browser, as the online version, and the settings above are the same. Some features of the paid versions are not included.

B.4 Tips to compile faster while writing.

- **Compile only the chapter you are editing.** In `main.tex` uncomment the line `\includeonly{chapters/chapter2}`. The other chapters are skipped but keep their numbers, citations and references.
- **Use Overleaf's Draft mode** (*Recompile* menu): it skips the images and is much faster.

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